

Aditya Kulkarni

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PROFESSIONAL SUMMARY

Backend engineer with production experience building distributed, scalable systems in Java and Python. Designed Kafka-driven event pipelines handling 2,000 async requests per second with Temporal fault-tolerant orchestration; built transactional Java 21 + Spring Boot APIs with PostgreSQL and MySQL data layers; and shipped 10+ backend features across 7 production deployments at Isomer AI. Strong in distributed systems design, SDLC end-to-end, and production observability with Prometheus and Grafana. MS CS at NC State (May 2026). Portfolio: kulkarniaditya.com

CORE COMPETENCIES

Java 21 + Spring Boot

Python + FastAPI

Apache Kafka

Distributed Systems

MySQL + PostgreSQL

Scalability Engineering

Prometheus + Grafana

Docker + CI/CD

WORK EXPERIENCE

NC State University Apr 2025 - Present

Graduate Research Assistant

- Designed a cloud-based genomic data platform with JBrowse 2 and MERN stack, enabling interactive exploration of 500GB strawberry and grape datasets for 100+ researchers worldwide, deployed on Google Cloud Run
- Collaborated on ML architecture design, applying hyperparameter tuning and time-series analysis to improve model accuracy by 12%; integrated Grafana dashboards for real-time performance monitoring

Isomer AI May 2025 - Present

Product Engineer Intern

- Built 10+ backend features for a production AI claims automation platform: Temporal.io workflow orchestration, real-time event pipelines (Pusher), and role-based access control across 7 production deployments -- 394 commits, full SDLC ownership in TypeScript, Node.js, PostgreSQL, and Redis
- Designed document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 file storage) and multi-model AI agent tooling across 3 providers and 8+ models (Claude, GPT-4o, Gemini) with Auth0, Docker, and Sentry observability

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows including a health claim automation tool using OCR, NLP, and ML; improved claim accuracy by 25% and automated AWS + GCP deployments with Terraform and Grafana monitoring dashboards

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Implemented distributed data pipelines using Hadoop + Spark for batch processing; integrated Kafka streaming for real-time data ingestion, reducing ticket-processing latency by 20% across a support ticket analysis system
- Led end-to-end SDLC for the data-driven system, reducing processing time by 15% through scalable pipeline design and distributed system architecture

PROJECTS

AlgoPulse -- Agentic AI Platform Distributed Systems

Architected an event-driven AI orchestration platform; identified API-polling bottleneck under load and pivoted to Supabase Edge Functions streaming AI reasoning loops directly to the client. Scaled worker architecture with Temporal.io and Kafka for fault-tolerant state synchronization -- 2,000 async inference requests per second without state loss.

Stack: TypeScript, Node.js, Supabase Edge Functions, Temporal.io, Kafka

Inventory Reservation Service Java + SQL

Built a transactional inventory reservation API in Java 21 + Spring Boot preventing overselling via PostgreSQL row-level locking (SELECT FOR UPDATE) across multi-SKU reservations; idempotent POST endpoints with RFC 9457 error responses and automated expiry via scheduled cleanup. Verified concurrency correctness with Testcontainers integration tests -- two concurrent reservations against quantity 1 produce exactly one success and one 409 Conflict.

Stack: Java 21, Spring Boot 3.5, PostgreSQL, Flyway, Spring Data JPA, Spring Security, Testcontainers, Docker

VectorLift -- Semantic Search and Ranking Engine Python + Kafka + Observability

Built a production-grade semantic search system on MS MARCO: BM25 baseline (Elasticsearch), dense bi-encoder retrieval (sentence-transformers fine-tuned with in-batch negatives), and cross-encoder re-ranker (MiniLM); evaluated across 6 pipeline configurations with NDCG@10, MRR@10, MAP. Designed a Kafka-driven indexing pipeline for event-based corpus ingestion; served search via FastAPI with per-stage latency tracking in Prometheus + Grafana.

Stack: Python, PyTorch, Elasticsearch, FAISS, Qdrant, Kafka, FastAPI, PostgreSQL, Redis, Prometheus, Grafana, Docker Compose

PulseRank -- Real-Time Leaderboard and Activity Feed MySQL + Redis + Scalability

Built an event-driven leaderboard platform: MySQL for transactional event writes with idempotency-key deduplication and outbox pattern, MongoDB for feed projections with TTL expiry, Redis sorted sets for sub-millisecond leaderboard reads; live score updates over Server-Sent Events. Benchmarked hot paths with k6; shipped observability with OpenTelemetry + Prometheus.

Stack: TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, OpenTelemetry, Prometheus, GitHub Actions

EDUCATION

North Carolina State University -- Master of Science, Computer Science	Aug 2024 - May 2026
Courses: Artificial Intelligence, Operating Systems, Data Science, Algorithms, Software Engineering, Object Oriented Development	
Mumbai University (DJSCE) -- Bachelor of Technology, Computer Science	Aug 2020 - May 2024
CGPA: 9.32/10 IEEE-published research on NLP document processing (T5 transformer, 1,500+ PDFs)	

SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, C **Backend:** Spring Boot, Node.js, FastAPI, NestJS, Kafka, Temporal, Express
Databases: PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch **Observability:** Prometheus, Grafana, Sentry, OpenTelemetry
Cloud + DevOps: AWS, GCP, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD
AI/ML: PyTorch, Sentence-Transformers, LangChain, OpenAI/Anthropic/Gemini APIs, RAG, FAISS, Qdrant